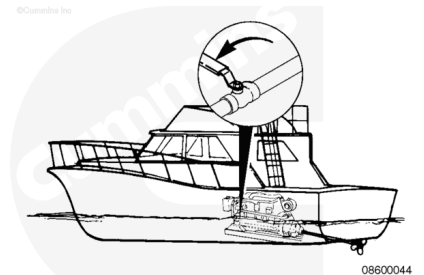


## Trainings ▾

### Remove

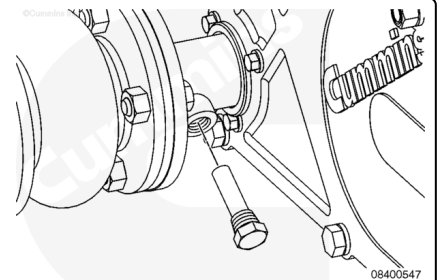
#### Marine Applications

Shut off the sea water inlet valve on the vessel hull.



#### ⚠ CAUTION ⚠

**In some cases, it will be necessary to hold the welded fitting on the heat exchanger with an additional wrench to prevent damage to the heat exchanger.**



Remove the zinc plug(s).

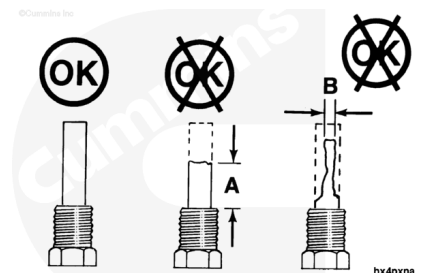
The illustration shown is generic. For engine-specific locations, Refer to Procedure 100-002 in Section E.  
(</qs3/pubsys2/xml/en/procedures/20/20-100-002-tr.html>)

### Inspect for Reuse

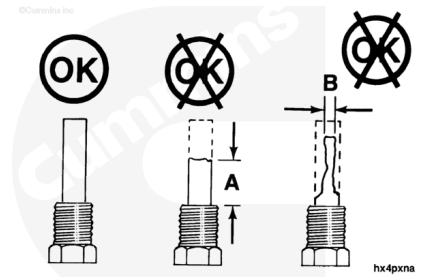
#### Marine Applications

Measure the length of the zinc plug(s) removed from the heat exchanger. Replace any zinc plug if it is 50 percent or more eroded.

**Note :** The frequency of change depends on the chemical reaction of the sea water circulated through the heat exchanger.



| Zinc Plug Erosion Limits |      |      |       |       |
|--------------------------|------|------|-------|-------|
|                          | mm   |      | in    |       |
| A                        | 19.0 | MIN  | 0.750 |       |
| New                      |      | 51.0 | MAX   | 2.000 |
| B                        | 6.4  | MIN  | 0.250 |       |
| New                      |      | 16.0 | MAX   | 0.630 |



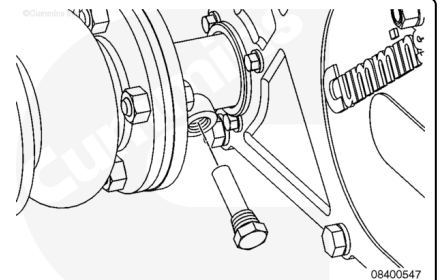
**Note :** If any zinc plug does not meet specification, replace it with a new zinc plug.

## Install

### Marine Applications

#### ⚠ CAUTION ⚠

**Do not use Teflon™ tape or pipe sealant on the threads of the sacrificial zinc anode plugs. Use of any type of sealant will insulate the zinc anode plugs and prevent a good ground, resulting in engine damage.**



Install and tighten the zinc plug(s).

#### Torque Value:

3.35 mm [0.25 in] Plug

1. 12 n•m [ 9 ft-lb ]

#### Torque Value:

9.525 mm [0.375 in] Plug

1. 22 n•m [ 17 ft-lb ]

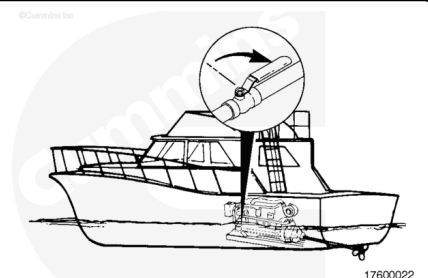
#### Torque Value:

12.7 mm [0.5 in] Plug

1. 30 n•m [ 22 ft-lb ]

Open the sea water valve on the vessel hull.

Operate the engine and check for leaks.



---

**Last Modified: 17-Sep-2015**

---